

## COMP433-S2-Group Project: Phase 4: System Modelling and Design

Tasks 4.1, 4.2: System Class and Sequence Design & Modelling

Tasks 4.3, 4.4, 4.5: System architectural and Component Design & Modelling

Tasks 4.6: System Deployment Design & Modelling

**Phase-4, Deadline: Monday 22<sup>th</sup> June** [complete the work of this phase on this deadline to be ready for the Presentation of all project phases, which will be on the dates below. [No submission is required for Phase-4, but SUBMIT tasks of this phase with the Final complete Project report submission, (see below)].

**FINAL Project Presentation: Wed 24<sup>th</sup> June (G1-G7), Saturday 27<sup>th</sup> June (G8-G13)**

**FINAL COMPLETE Report SUBMISSION Deadline: Tuesday 23<sup>th</sup> June (see below for details)**

Dear Students,

After the completion of Requirement Discovery, Analysis and Modelling phases and have produced final use case and activity models, now you are required to undertake the next phase of your project: **Phase 4: System Modelling and Design**. In this phase, each group is required to undertake the following tasks, as described below.

### **Task 4.1- [Group Task] System Class modelling and Analysis:**

1. Draw an ANALYSIS CLASS Diagram.
2. Draw a DETAILED CLASS Diagram, inclusive of data attributes and operations/methods, for each of the classes.

### **Task 4.2- [Individual Task] System Sequence modelling and Analysis:**

Draw **Four** or **Five** SEQUENCE diagrams: **Each member** of the group shall draw **one** sequence diagram, one for each use-case, the same use-case that the member of the group has written a detailed use-case description for.

### **Task 4.3- [Group Task] System Design Goals**

Consider and describe clearly at least three design goals, **TWO** general design goals, e.g. low couple, high cohesion, and at least **ONE** specific system design goal, derived from your business non-functional requirements (, e.g. performance in terms of processing speed; reliability e.g. in terms of number of faults it can make in a day; user-friendliness e.g. in terms of training days;) or non-functional constraints, e.g. has to run on MS windows, or has to work under low temperature in the north pole. Describe these three design goals clearly.

### **Task 4.4- [Group Task] System and Component modelling**

Draw, **Component diagram** of your chosen System design, taking into account the THREE design goals above.

#### **Task 4.5- [Group Task] System and architectural Design**

Draw, your chosen architectural design of your system according to the above THREE design goals using, **Component diagram** or an **Architecture diagram**.

#### **Task 4.6- [Group Task] System and Deployment modelling**

Draw a **deployment diagram** of your system, by mapping your software components on potential hardware nodes, taking into account the THREE design goals above.

=> **SUBMISSION:** Submit final versions of phase-4 tasks, **with the FINAL COMPLETE PROJECT REPORT** (I will send further details later).

=> **FINAL Project Presentation** of all your project work/phases will be on **the above dates** during the lecture. You are required to submit a copy of your project presentation, inclusive of photos of group members, with the Final Project Report. The format of the Final Project Presentation is specified below.

I will send further details, in a separate email, on the format and requirements of the FINAL COMPLETE PROJECT REPORT submission.

#### **Format of Final Group Project Presentations**

As confirmed previously, the final presentation of your project will be on **the above dates** during the lecture. Please come on time for the lecture, groups who come late or lose time in getting their presentation ready, lost time will be deducted from their allocated presentation time.

**Groups** are required to present their ALL-project work, fully. Groups are required to present the following parts of your project, as follows:

- 1- Group and its members and their roles – 1 slide
- 2- Business description/introduction (briefly) – 1 slide
- 3- Overall Use case diagram - 1 slide
- 4- One example of a USE CASE description/specification
- 5- Overall Activity diagram – 1 slide
- 6- Overall Analysis Class Model/Diagram - 1 slide
- 7- Overall (detailed) Class Model/Diagram – 1 slide
- 8- One Sequence diagram, of the same Use Case, in point 3 above – 1 slide
- 9- Design Goals – 1 slide
- 10- Overall architecture (using architecture diagram) – 1 slide
- 11- Component diagram – 1 slide
- 12- Deployment diagram – 1 slide

Presentations are an essential and major part of your project assessment and therefore, all members of the group should be involved in the presentation. Whoever, group or member of

a group, is absent will receive ZERO mark for this part of the project (and obviously most probably will fail the project assessment). Attendance will be as normal and thus all students are expected to attend.

All members of the group should be involved in the presentation, with balanced/ equal time.

**Each Group will have 12 min to present + 3 min for discussion.** Time is thus very tight, and will be strictly followed, students will be stopped on time -thus each group need to practice to ensure they do not exceed their allocated time. Also, to avoid delays, you are required to have your presentation ready, e.g., in PowerPoint. Lost time will be deducted from the group's time. If a group fails to have their presentation ready in time, it will be treated as if a group has not presented.

Groups will be asked questions and will receive feedback on their work and will be asked to reflect the feedback onto their final project report submission.